

Question number	Answer	Additional guidance	Mark
6(a)(i)	An explanation that includes the following points: • because both lines rise and fall (in parallel) (1) • but the line for deaths from lung cancer is a period of time after the line for cigarettes smoked (1)	Accept {delay / timelag} for deaths	(2)

Question number	Answer	Additional guidance	Mark
6(a)(ii)	An explanation that includes the following points: • alveoli will have a smaller surface area (to volume ratio) (1) • therefore the (rate of) diffusion of oxygen into the bloodstream will be slower (1)	Accept gas exchange area will be smaller Accept gas exchange for diffusion Ignore less gas exchange / diffusion	(2)

Question number	Answer						
*6(b)	Indicative content: <table border="1" data-bbox="456 336 2063 1377"> <tr> <td data-bbox="456 336 792 699">Egg points (E)</td><td data-bbox="792 336 2063 699"> 1. {number / density of pores} 2. {width / size} of pores 3. area of pores 4. thickness of shell and membranes 5. rate of respiration of developing embryo 6. rate of blood flow of embryo 7. temperature 8. shell is impermeable / pores are permeable 9. shell supports membrane so there is a large SA for gas exchange </td></tr> <tr> <td data-bbox="456 699 792 1102">Theory points (T)</td><td data-bbox="792 699 2063 1102"> 10. Ficks Law of diffusion can be used to calculate diffusion rate 11. State Fick's Law (3) 12. because rate of diffusion depends on surface area / mp11 13. because rate of diffusion depends on diffusion distance / mp11 14. because rate of diffusion depends on concentration gradient / mp11 15. speed of molecules depends on temperature 16. because rate of diffusion depends on what substances oxygen is passing through eg water / air 17. diffusion coefficient through air 18. diffusion coefficient through membranes </td></tr> <tr> <td data-bbox="456 1102 792 1377">Explanation points (X) Accept converse for all</td><td data-bbox="792 1102 2063 1377"> 19. increasing SA (or any named factor increasing this) causes increase in rate of diffusion 20. increasing distance (or any named factor increasing this) causes decrease in rate of diffusion 21. increasing conc gradient (or any named factor increasing this) causes increase in rate of diffusion 22. increasing temperature increases rate of diffusion </td></tr> </table>	Egg points (E)	1. {number / density of pores} 2. {width / size} of pores 3. area of pores 4. thickness of shell and membranes 5. rate of respiration of developing embryo 6. rate of blood flow of embryo 7. temperature 8. shell is impermeable / pores are permeable 9. shell supports membrane so there is a large SA for gas exchange	Theory points (T)	10. Ficks Law of diffusion can be used to calculate diffusion rate 11. State Fick's Law (3) 12. because rate of diffusion depends on surface area / mp11 13. because rate of diffusion depends on diffusion distance / mp11 14. because rate of diffusion depends on concentration gradient / mp11 15. speed of molecules depends on temperature 16. because rate of diffusion depends on what substances oxygen is passing through eg water / air 17. diffusion coefficient through air 18. diffusion coefficient through membranes	Explanation points (X) Accept converse for all	19. increasing SA (or any named factor increasing this) causes increase in rate of diffusion 20. increasing distance (or any named factor increasing this) causes decrease in rate of diffusion 21. increasing conc gradient (or any named factor increasing this) causes increase in rate of diffusion 22. increasing temperature increases rate of diffusion
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	Level 1 Up to 4 points from anywhere; 2 points for one mark and 4 points for 2 marks Level 2 5 points or more, from two best categories; 5 points for 3 marks, 6 points for 4 marks. Must achieve 6 marks to progress to level 3 Level 3 As level 2 plus up to 2 points from third category; 7 points for 5 marks and 8 points for 6 marks
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